Review of Last Lecture

3.7 Basic Properties of Fourier Transform

3.7.1 Linearity

if $\mathcal{F}[f_1(t)] = F_1(\omega),\ \mathcal{F}[f_2(t)] = F_2(\omega),$

then $\mathcal{F}[a_1 f_1(t) + a_2 f_2(t)] = a_1 F_1(\omega) + a_2 F_2(\omega)$

3.7.2 Symmetry

if $\mathcal{F}[f(t)] = F(\omega)$ **then** $\mathcal{F}[F(t)] = 2\pi f(-\omega)$

$$f(t) \leftrightarrow F(\omega)$$

$$\frac{F(t)}{2\pi} \leftrightarrow f(\omega) \quad (f \text{ is an even function})$$

$$\frac{-F(t)}{2\pi} \leftrightarrow f(\omega) \quad (f \text{ is an odd function})$$

$$f(t) \leftrightarrow F(\omega)$$

$$F(t) \leftrightarrow 2\pi f(\omega) \quad (f \text{ is an even function})$$

$$F(t) \leftrightarrow -2\pi f(\omega) \quad (f \text{ is an odd function})$$

3.7.3 Parity, Realness, and Imaginariness

$$F(\omega)=\int_{-\infty}^{\infty} f(t)e^{-j\omega t}dt=\int_{-\infty}^{\infty} f(t)\cos(\omega t)dt-j\int_{-\infty}^{\infty} f(t)\sin(\omega t)dt=R(\omega)+jX(\omega)$$

$$\begin{cases} R(\omega)=\int_{-\infty}^{\infty} f(t)\cos(\omega t)dt \\ X(\omega)=-\int_{-\infty}^{\infty} f(t)\sin(\omega t)dt \end{cases}$$

If $f(t)$ is a real-valued function, then:

- **The real part of $F(\omega)$ $\left(R(\omega)\right)$ is even;**
- **The imaginary part of $F(\omega)$ $\left(X(\omega)\right)$ is odd.**

Case 1: $f(\mathrm{t})$ is real and even, $F(\omega)=R(\omega)$ (real, even).

Case 2: $f(\mathrm{t})$ is real and odd, $F(\omega)=jX(\omega)$ (purely imaginary, odd).

$$\left|F(\omega)\right|=\sqrt{R^2(\omega)+X^2(\omega)},\ \ \varphi(\omega)=\arctan\frac{X(\omega)}{R(\omega)}$$

If f(t) is real or purely imaginary $(f(t)=jg(t),\ g(t)$ real), then $\left|F(\omega)\right|$ is even, and $\varphi(\omega)$ is odd.

3.7.4 Shift Property

(1) Time-Shift Property

If $F(\omega) = \mathcal{F}[f(t)]$ **Then** $\mathcal{F}[f(t-t_0)] = F(\omega)e^{-j\omega t_0} = |F(\omega)|e^{j\varphi(\omega)}e^{-j\omega t_0}$

Amplitude spectrum remains unchanged. Phase spectrum has a phase shift of -ωt_0.

Similarly, $\mathcal{F}[f(t+t_0)] = F(\omega)e^{j\omega t_0}$

(2) Frequency-Shift Property (Modulation Principle)

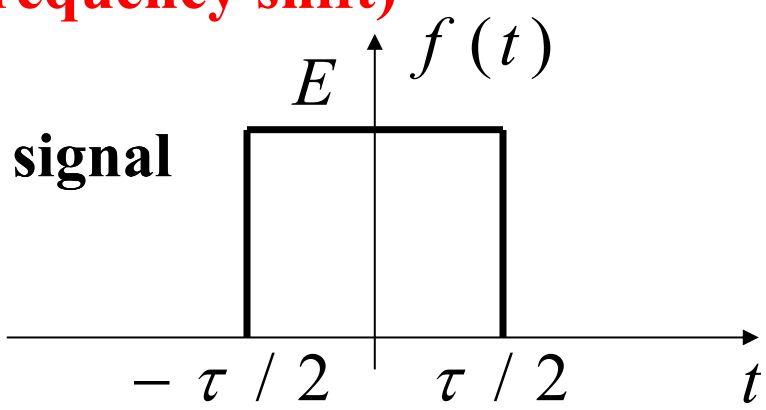
If $\mathcal{F}[f(t)] = F(\omega)$, then $\mathcal{F}\left[f(t)e^{j\omega_0 t}\right] = F(\omega-\omega_0)$, $\mathcal{F}\left[f(t)e^{-j\omega_0 t}\right] = F(\omega+\omega_0)$

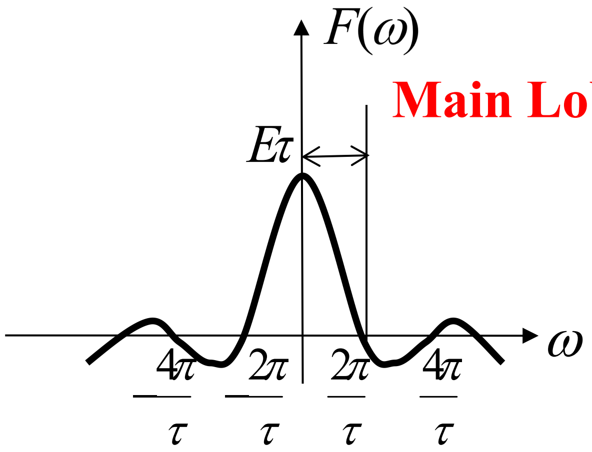
$$\mathcal{F}[f(t)\cos\omega_0 t] = \frac{1}{2}[F(\omega+\omega_0)+F(\omega-\omega_0)] \quad \mathcal{F}[f(t)\sin\omega_0 t] = \frac{j}{2}[F(\omega+\omega_0)-F(\omega-\omega_0)]$$

3.7 Basic Properties of Fourier Transform

Modulation (frequency shift)

Baseband signal

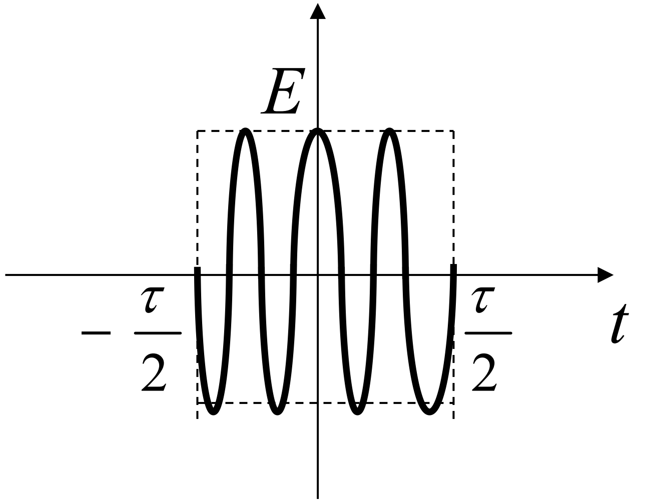




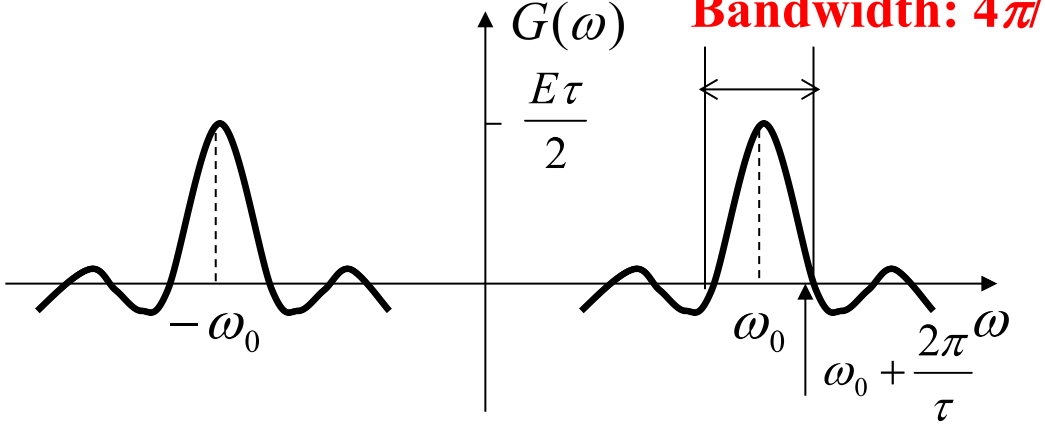
Main Lobe Bandwidth: $2\pi/\tau$

$$F(\omega) = E\tau \mathrm{Sa}(\frac{\omega\tau}{2})$$

ASK signal $g(t) = f(t)\cos(\omega_0 t)$



Main Lobe Bandwidth: $4\pi/\tau$



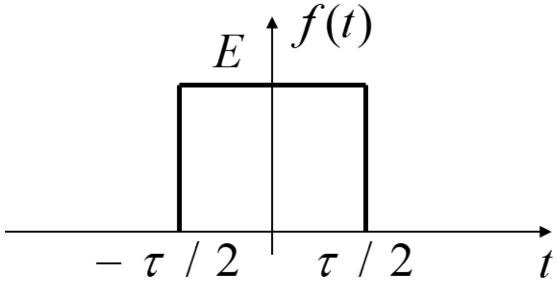
$$G(\omega) = \frac{E\tau}{2}\{\mathrm{Sa}[(\omega+\omega_0)\frac{\tau}{2}] + \mathrm{Sa}[(\omega-\omega_0)\frac{\tau}{2}]\}$$

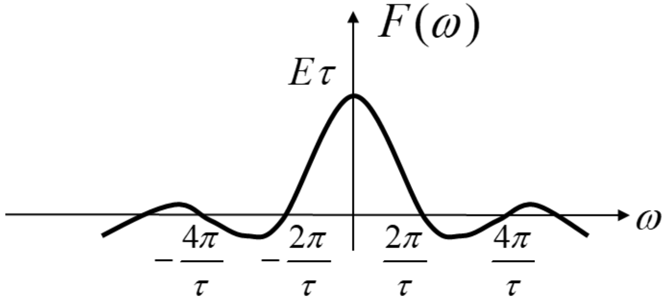
During modulation, (1) bandwidth doubles; (2) spectral density halves.

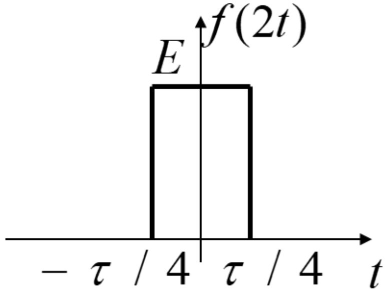
3.7.5 Scaling Property

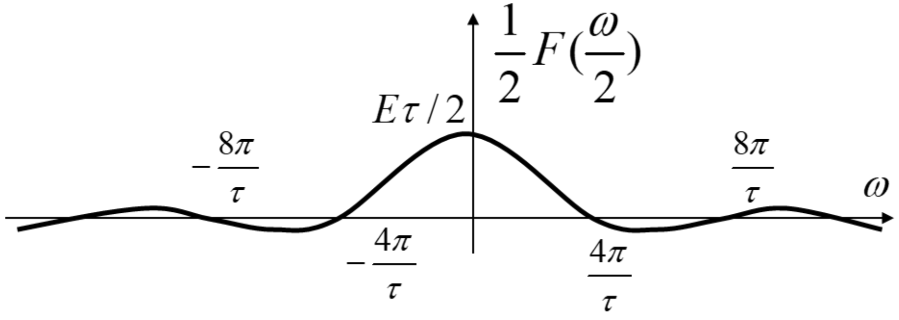
If $F(\omega) = \mathcal{F}[f(t)]$ **Then** $\mathcal{F}[f(at)] = \frac{1}{|a|}F(\frac{\omega}{a})$

Compressing a signal in time domain by *a* (*a*>1) times (signal changes faster) is equivalent to expanding the spectrum by *a* times in frequency domain (more high-frequency components) and reducing the spectral amplitude by 1/*a* (energy conservation). Similarly, time expansion is equivalent to frequency compression.









Special case: $\mathcal{F}[f(-t)] = F(-\omega)$

（**time reversal, *a* = -1**）

Combined Property:

Scaling + Time Shift

$$\boxed{\mathcal{F}[f(at-t_0)] = \frac{1}{|a|}F(\frac{\omega}{a})e^{-j\frac{\omega t_0}{a}}}$$

3.7.6 Differentiation and Integration Properties

(1) Time-Domain Differentiation Property

If $F(\omega) = \mathcal{F}[f(t)]$

Then $\mathcal{F}\left[\dfrac{df(t)}{dt}\right] = j\omega F(\omega)$, $\quad \mathcal{F}\left[\dfrac{d^n f(t)}{dt^n}\right] = (j\omega)^n F(\omega)$

(2) Time-Domain Integration Property

If $F(\omega) = \mathcal{F}[f(t)]$

Then $\mathcal{F}\left[\int_{-\infty}^{t} f(\tau)d\tau\right] = \dfrac{F(\omega)}{j\omega} + \pi F(0)\delta(\omega)$

DC Component

(3) Frequency-Domain Differentiation Property

If $F(\omega) = \mathcal{F}[f(t)]$

Then $$\boxed{\mathcal{F}[t^n f(t)] = j^n \frac{d^n F(\omega)}{d\omega^n}}$$

(4) Frequency-Domain Integration Property

If $F(\omega) = \mathcal{F}[f(t)]$

Then $$\mathcal{F}^{-1}\left[\int_{-\infty}^{\omega} F(u)du\right] = \frac{f(t)}{-jt} + \pi f(0)\delta(t)$$

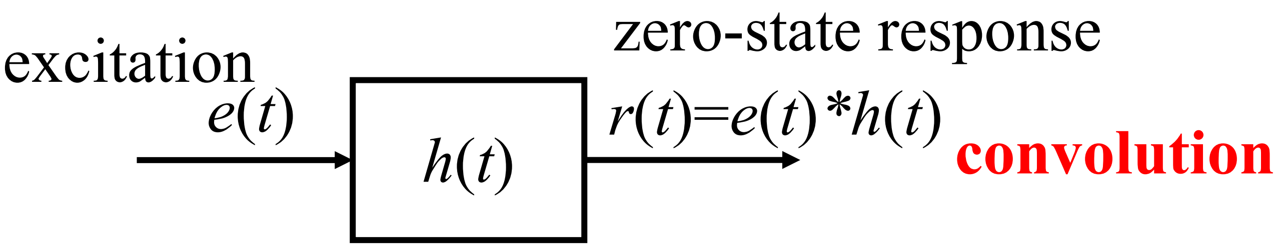
Contents of This Lecture

Learning Objectives of This Lecture

1. Master proficiently the **Time-domain Convolution Theorem**, the **Frequency-domain Convolution Theorem** and their applications;
2. Be familiar with the relationship between the **Fourier Transform of periodic signals** and the Fourier Series;
3. Master proficiently the **Sampling Theorem** and its applications.

Review of Convolution

Convolution is a core technique in system analysis. The **zero-state response** of a system is the **convolution** of the excitation and the **impulse response**.



- **Four Steps** of Convolution Operation:

1. Time Reversal: $h(\tau) \rightarrow h(-\tau)$ **2. Time Shifting:** $h(-\tau) \rightarrow h(t-\tau)\ \begin{cases} t<0, & \text{left shift by } t \\ t>0, & \text{right shift by } t \end{cases}$

3. Multiplication: $e(\tau)h(t-\tau)$ **4. Integration:** $e(t)*h(t) = \int_{-\infty}^{\infty} e(\tau)h(t-\tau)d\tau$ Computation is complex!

- **Convolution Theorem**—**Convert complex time-domain computations into simple frequency-domain computations, and vice versa**.
- **Sampling Theorem**—An application of the Convolution Theorem.

Selection of Sampling Frequency

To convert a time-continuous piece of music into digital music in a CD, it is necessary to perform sampling (time discretization) on the analog signal. Which of the following sampling frequencies (unit: Hz, representing the number of samples per second) do you think achieves the highest music quality?

44 kHz Sampling Frequency

22 kHz Sampling Frequency

11 kHz Sampling Frequency

5.5 kHz Sampling Frequency

3.8 Convolution Theorem

(1) Time-domain Convolution Theorem

The time-domain convolution of two signals is equivalent to the multiplication of their spectra in the frequency domain.

if $F_1(\omega) = \mathcal{F}[f_1(t)]$, $\quad F_2(\omega) = \mathcal{F}[f_2(t)]$,

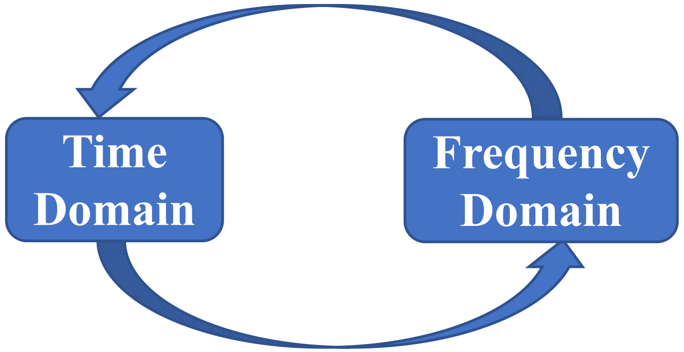
then $$\boxed{\mathcal{F}\,[f_1(t) * f_2(t)] = F_1(\omega) F_2(\omega)}$$

(2) Frequency-domain Convolution Theorem

The time-domain convolution of two signals is equivalent to the multiplication of their spectra in the frequency domain.

if $F_1(\omega) = \mathcal{F}[f_1(t)]$, $\quad F_2(\omega) = \mathcal{F}[f_2(t)]$,

then $$\boxed{\mathcal{F}\,[f_1(t) \cdot f_2(t)] = \frac{1}{2\pi} F_1(\omega) * F_2(\omega)}$$



where: $$F_1(\omega) * F_2(\omega) = \int_{-\infty}^{\infty} F_1(u) F_2(\omega - u)\, du$$

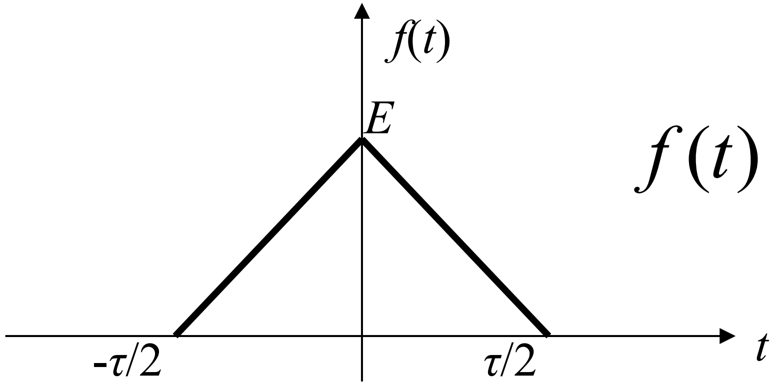
Example 3-7: Use the Time-domain Convolution Theorem to find the spectrum of a triangular pulse:

$$f(t)=\begin{cases} E(1-\frac{2|t|}{\tau}) & |t|\le\frac{\tau}{2} \\ 0 & |t|>\frac{\tau}{2} \end{cases}$$

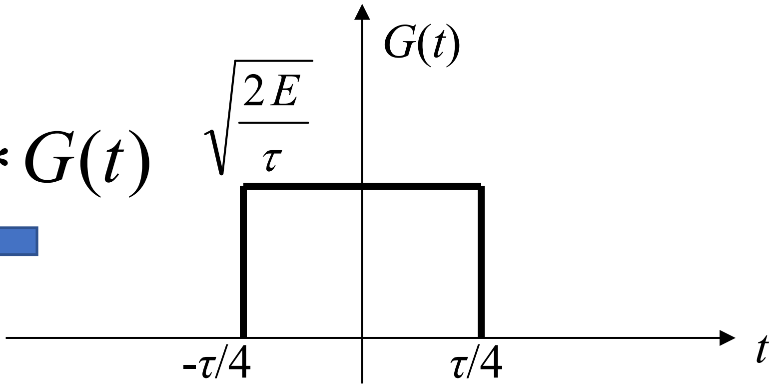
Solution:

Step 1: Treat the triangular pulse as the convolution of two identical rectangular pulses.

The width of the triangular pulse is the sum of the widths of two rectangular pulses. The peak amplitude of the triangular pulse is the product of the square of the amplitudes of the two rectangular pulses and their width. Therefore, the width and amplitude of each rectangular pulse are $\tau/2$ and $\sqrt{2E/\tau}$, respectively.



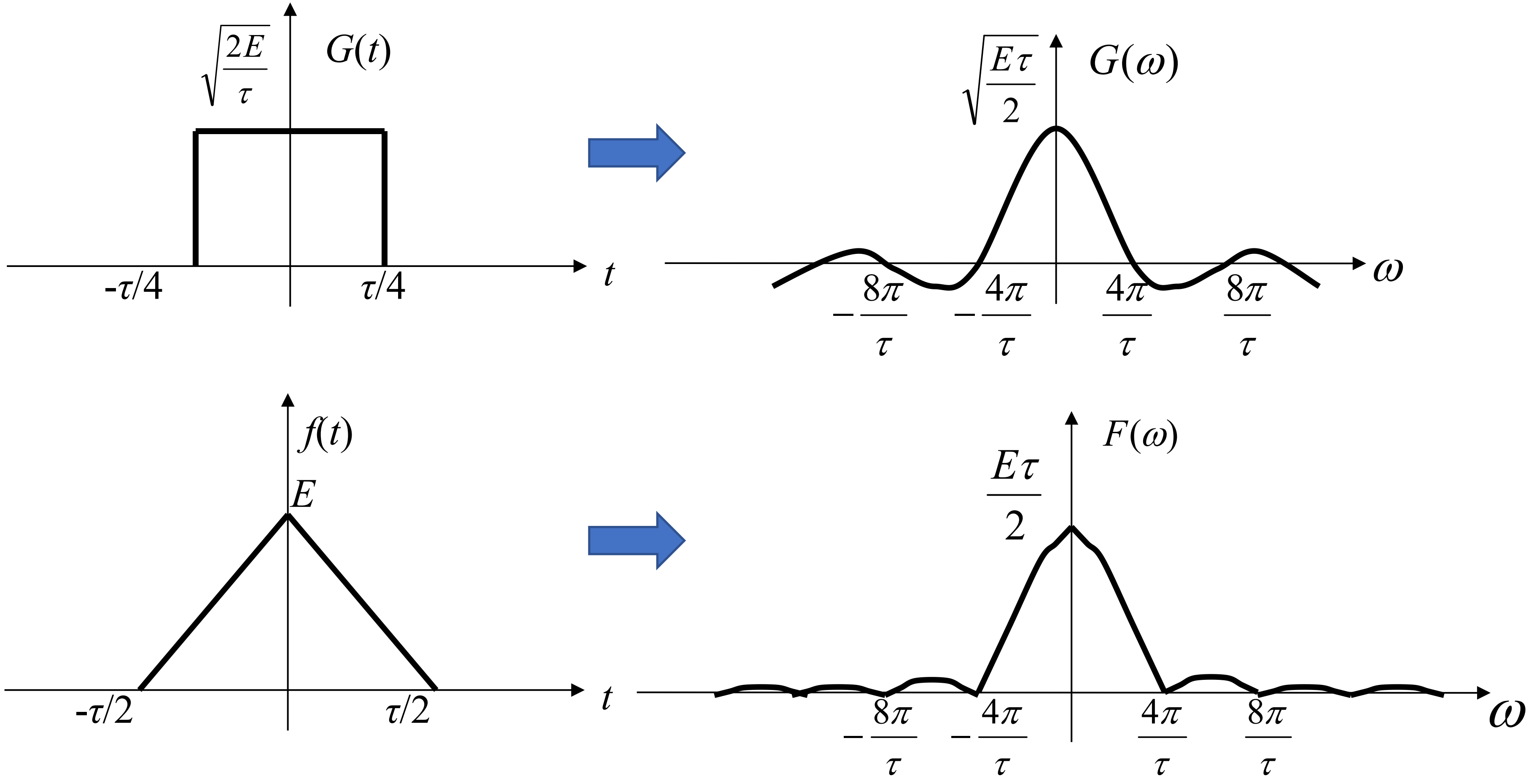
$$f(t)=G(t)*G(t)$$



3.8 Convolution Theorem

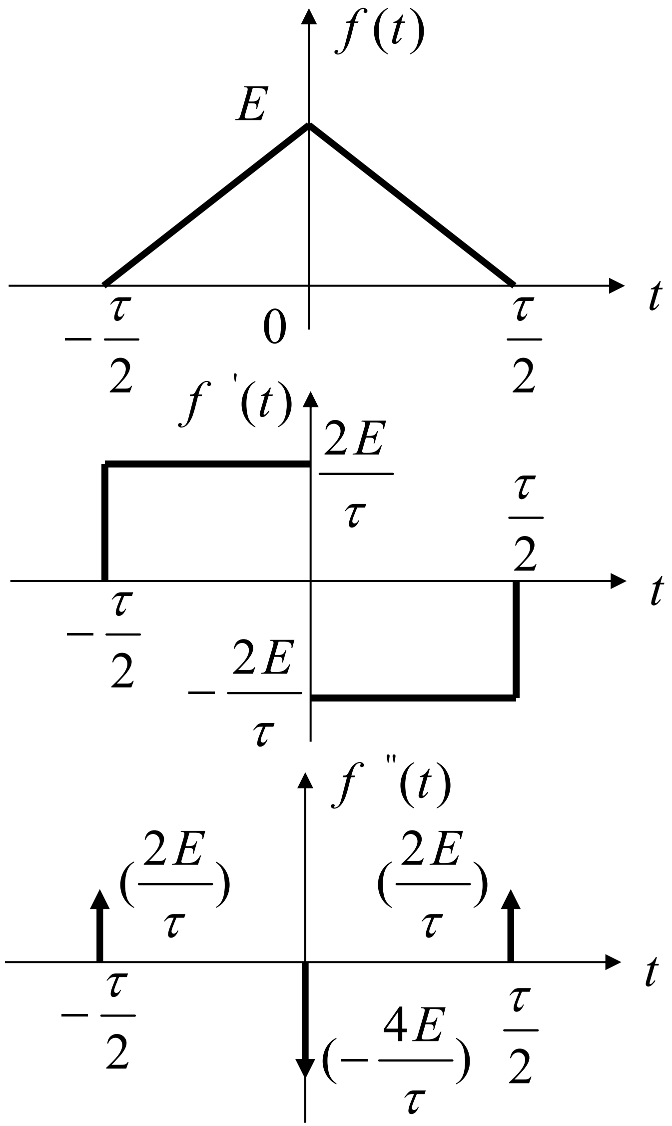
Step 2: Apply the Time-domain Convolution Theorem.

$$G(\omega)=\sqrt{\frac{2E}{\tau}}\cdot\frac{\tau}{2}\mathrm{Sa}\left(\frac{\omega\tau}{4}\right)=\sqrt{\frac{E\tau}{2}}\mathrm{Sa}\left(\frac{\omega\tau}{4}\right)\qquad\therefore\quad F(\omega)=G^2(\omega)=\frac{E\tau}{2}\mathrm{Sa}^2\left(\frac{\omega\tau}{4}\right)$$



3.8 Convolution Theorem

Review **Example 3-6:** Use **the Time-domain Differentiation Property** to find the Fourier Transform of the Triangular Pulse Signal.



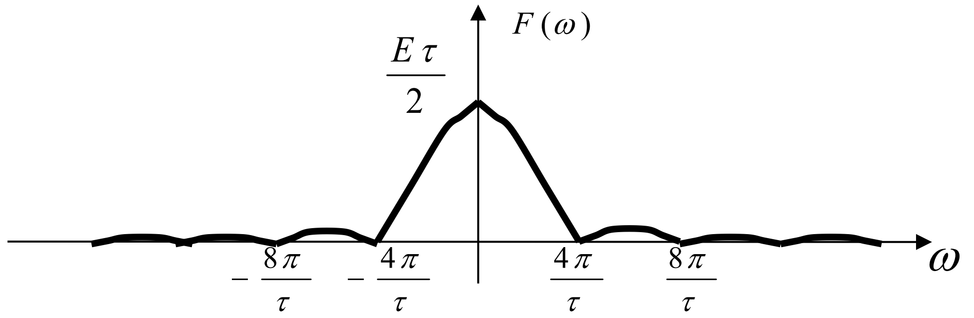
Solution: $f''(t) = \frac{2E}{\tau}[\delta(t+\frac{\tau}{2}) + \delta(t-\frac{\tau}{2}) - 2\delta(t)]$

Take the Fourier Transform on both sides of the above equation:

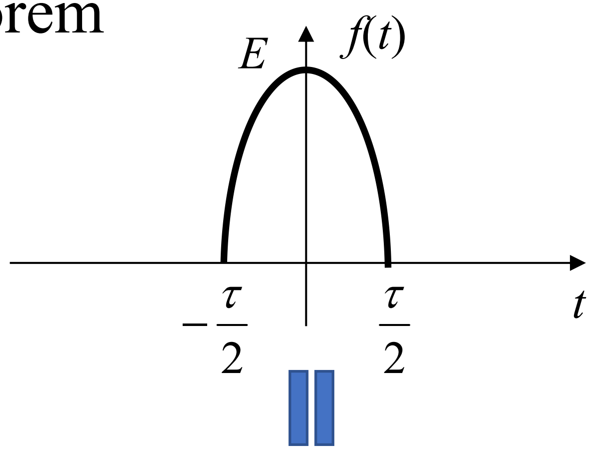
$$(j\omega)^2 F(\omega) = \frac{2E}{\tau}(e^{j\omega\frac{\tau}{2}} + e^{-j\omega\frac{\tau}{2}} - 2)$$

$$= -\frac{\omega^2 E\tau}{2}\mathrm{Sa}^2(\frac{\omega\tau}{4})$$

$$\therefore \quad F(\omega) = \frac{E\tau}{2}\mathrm{Sa}^2(\frac{\omega\tau}{4})$$



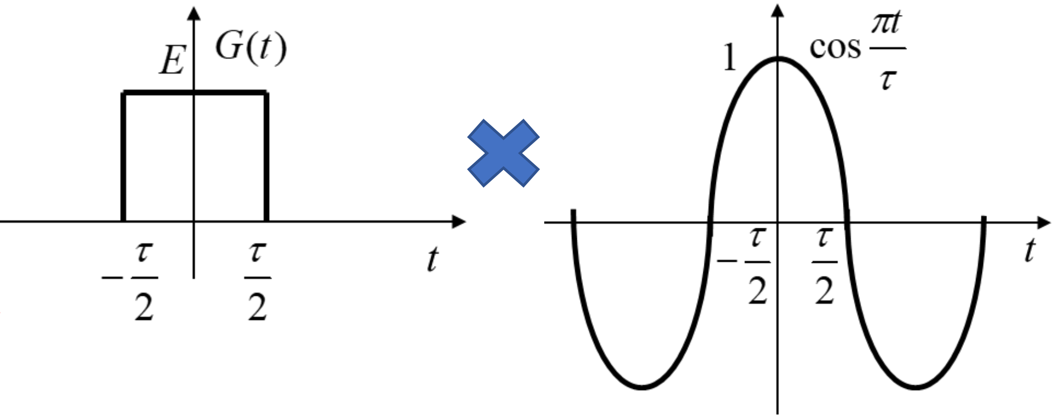
Example 3-7: Use the Frequency-domain Convolution Theorem to Find the Spectrum of a Cosine Pulse.



$$f(t)=\begin{cases} E\cos\dfrac{\pi t}{\tau} & |t|\le\dfrac{\tau}{2} \\ 0 & |t|>\dfrac{\tau}{2} \end{cases}$$

Solution:

Step 1: Treat $f(t)$ as the product of a rectangular pulse $G(t)$ and a cosine signal.



Step 2: Apply the Frequency-domain Convolution Theorem.

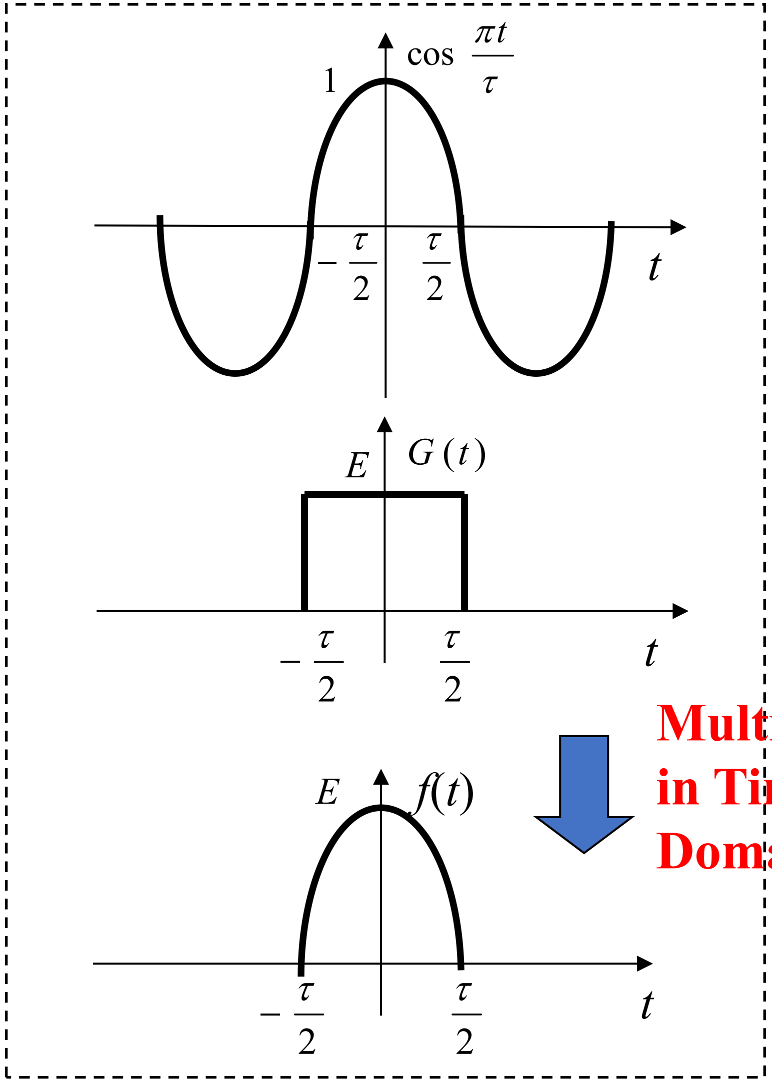
$$G(\omega)=E\tau\mathrm{Sa}\left(\frac{\omega\tau}{2}\right)$$

$$\mathcal{F}\left[\cos\frac{\pi t}{\tau}\right]=\pi\delta\left(\omega+\frac{\pi}{\tau}\right)+\pi\delta\left(\omega-\frac{\pi}{\tau}\right)$$

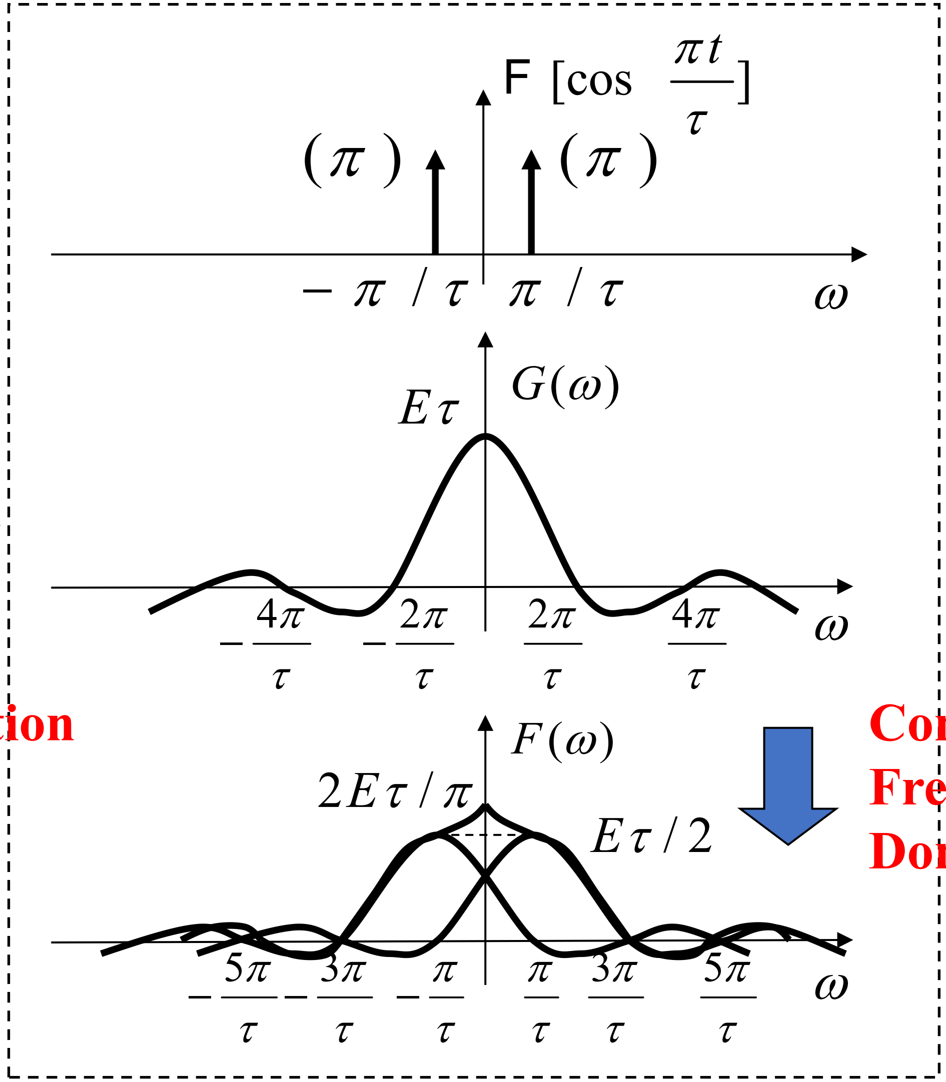
$$F(\omega)=\frac{1}{2\pi}G(\omega)*\pi\left[\delta\left(\omega+\frac{\pi}{\tau}\right)+\delta\left(\omega-\frac{\pi}{\tau}\right)\right]=\frac{E\tau}{2}\left\{\mathrm{Sa}\left[\left(\omega+\frac{\pi}{\tau}\right)\frac{\tau}{2}\right]+\mathrm{Sa}\left[\left(\omega-\frac{\pi}{\tau}\right)\frac{\tau}{2}\right]\right\}$$

3.8 Convolution Theorem

$$F(\omega) = \frac{E\tau}{2}\{\mathrm{Sa}[(\omega+\frac{\pi}{\tau})\frac{\tau}{2}] + \mathrm{Sa}[(\omega-\frac{\pi}{\tau})\frac{\tau}{2}]\} = 2E\tau\cos(\frac{\omega\tau}{2}) \Big/ \left[\pi\left(1-\frac{\tau^2\omega^2}{\pi^2}\right)\right]$$

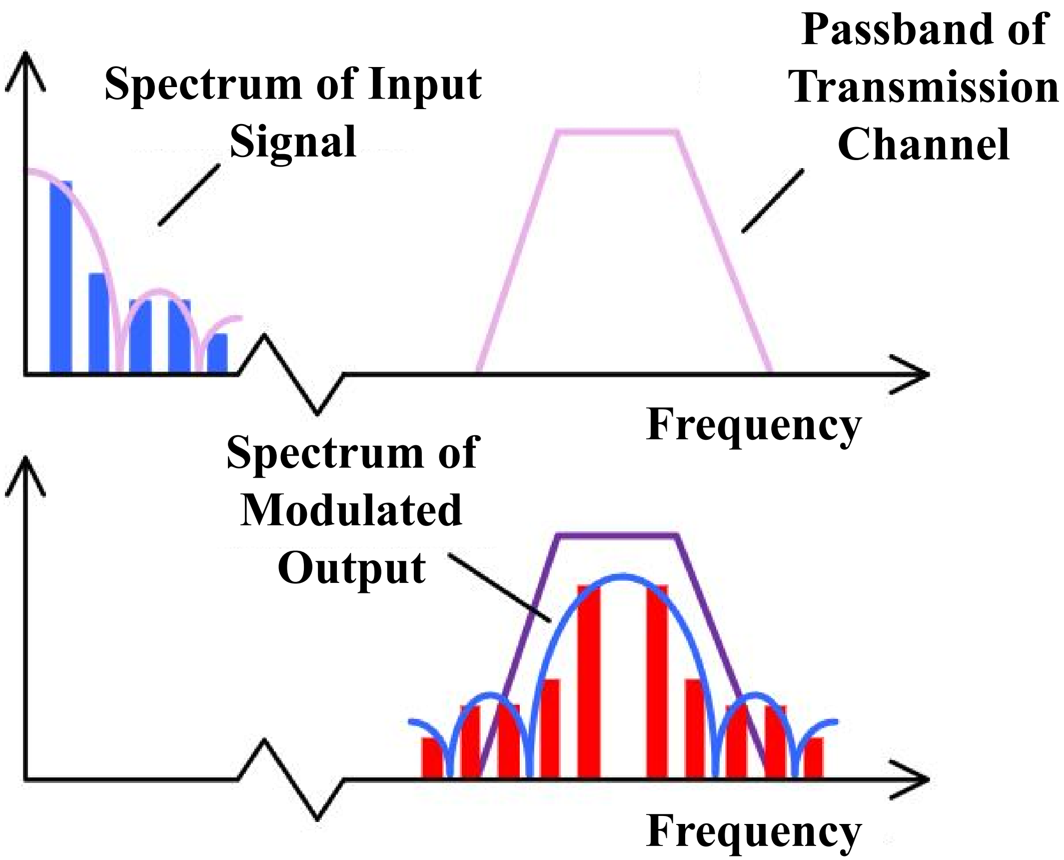


Multiplication in Time Domain



Convolution in Frequency Domain

Application of the Time-domain Convolution Theorem in Communications



$$x(t) * h(t) \xrightarrow{F} X(\omega)H(\omega)$$

Time-domain Convolution Theorem

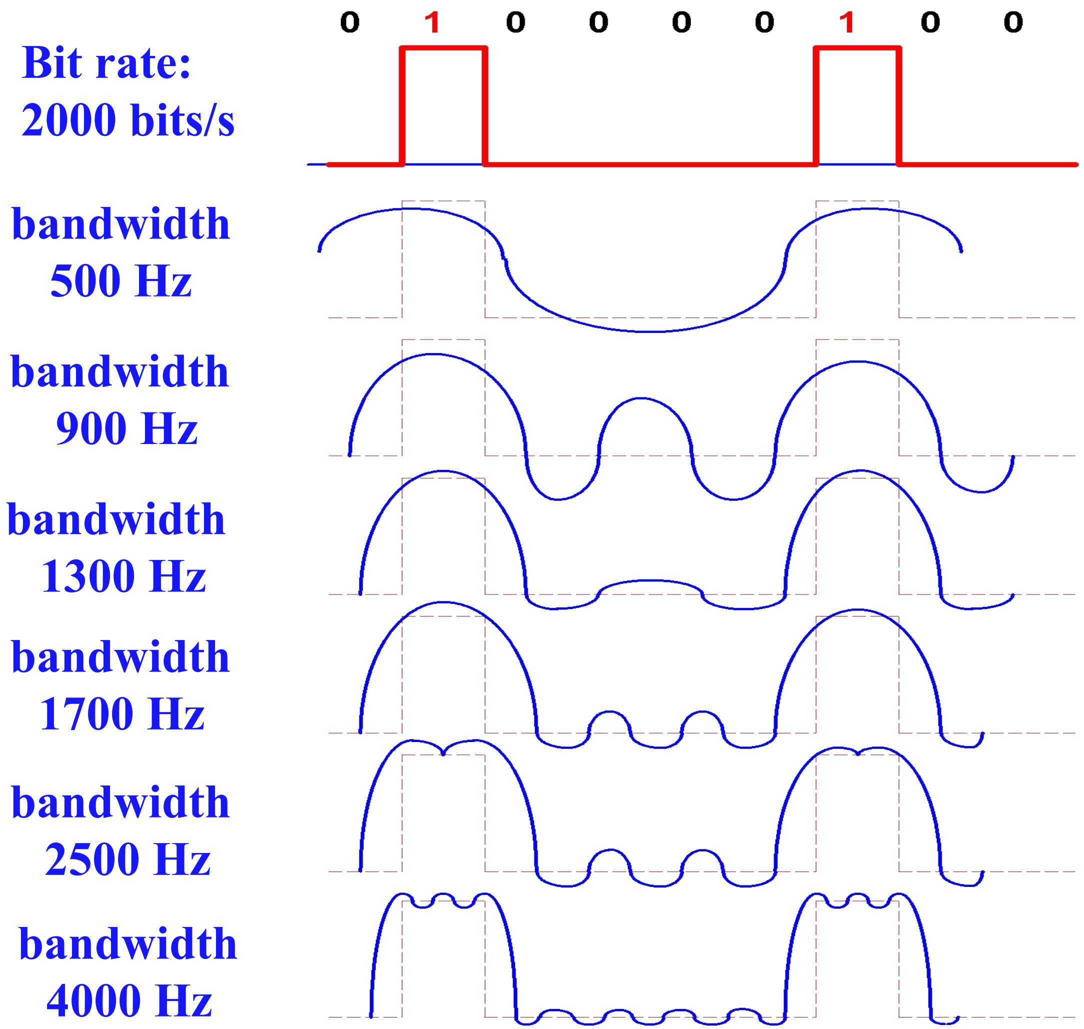
- The channel bandwidth is limited.
- If some main frequency components of the signal fall outside the channel bandwidth, the signal output by the channel will be distorted.

3.8 Convolution Theorem

Application of the Time-domain Convolution Theorem in Communications

- Signal bandwidth: 2000 Hz

- Output signals of the channel under different bandwidths:



3.9 Fourier Transform of Periodic Signals

$$\begin{tikzcd}[column sep=huge, row sep=huge]
\textbf{Periodic Signal} \arrow[r] \arrow[d, "T_1 \to \infty"] \arrow[dr, "?"] & \textbf{Fourier Series} \arrow[d, "T_1 \to \infty,\ \times T_1"] \\
\textbf{Aperiodic Signal} \arrow[r] & \textbf{Fourier Transform}
\end{tikzcd}$$

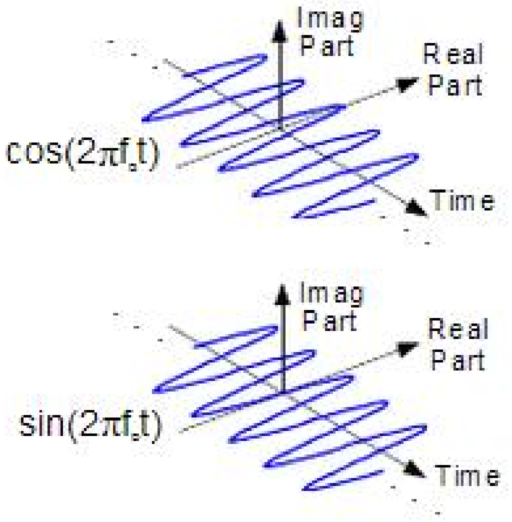
(represents spectral density)

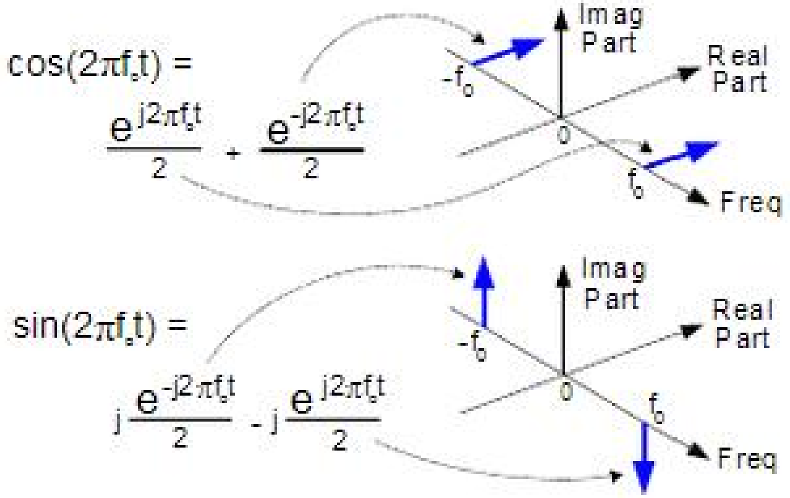
3.9 Fourier Transform of Periodic Signals

3.9.1 Fourier Transform of Sine and Cosine Signals

$$\cos(\omega_0 t) = \frac{1}{2}\left(e^{j\omega_0 t} + e^{-j\omega_0 t}\right)$$

$$\sin(\omega_0 t) = \frac{1}{2j}\left(e^{j\omega_0 t} - e^{-j\omega_0 t}\right)$$





$$\mathrm{F}\left[e^{j\omega_0 t}\right] = 2\pi\delta(\omega - \omega_0)$$

$$\mathrm{F}\left[e^{-j\omega_0 t}\right] = 2\pi\delta(\omega + \omega_0)$$

$$\mathrm{F}\left[\cos\omega_0 t\right] = \pi[\delta(\omega + \omega_0) + \delta(\omega - \omega_0)]$$

$$\mathrm{F}\left[\sin\omega_0 t\right] = j\pi[\delta(\omega + \omega_0) - \delta(\omega - \omega_0)]$$

3.9.2 Fourier Transform of General Periodic Signals

Let the period of the periodic signal $f(t)$ be T_1 , and the angular frequency be $\omega_1 = 2\pi f_1 = 2\pi / T_1$. Its Fourier series is:

$$f(t) = \sum_{-\infty}^{\infty} F_n e^{jn\omega_1 t} \qquad \textbf{(1)}$$

where: $F_n = \dfrac{1}{T_1}\int_{-T_1/2}^{T_1/2} f(t) e^{-jn\omega_1 t} dt$

Take the Fourier transform on both sides of equation (1):

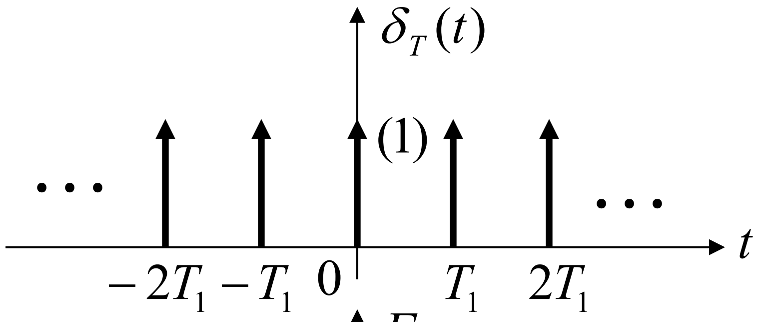
$$F(\omega) = \mathcal{F}[f(t)] = \mathcal{F}\left[\sum_{n=-\infty}^{\infty} F_n e^{jn\omega_1 t}\right] = \sum_{n=-\infty}^{\infty} F_n \mathcal{F}\left[e^{jn\omega_1 t}\right] = 2\pi \sum_{n=-\infty}^{\infty} F_n \underline{\delta(\omega - n\omega_1)}$$

Using the Frequency-shift Property

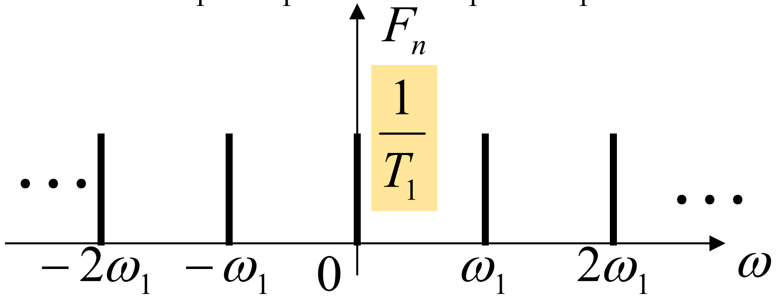
that is, $\boxed{F(\omega) = 2\pi \sum_{n=-\infty}^{\infty} F_n \delta(\omega - n\omega_1)}$

The Fourier transform of the periodic signal $f(t)$ is composed of a series of impulse functions, which are located at the harmonic frequencies of the signal $(0, \pm\omega_1, \pm 2\omega_1, \cdots\cdots)$ The strength of each impulse is equal to 2π times the corresponding coefficient F_n of the Fourier series of $f(t)$.

Example 3-9: Find the Fourier Transform of the Periodic Unit Impulse Sequence.



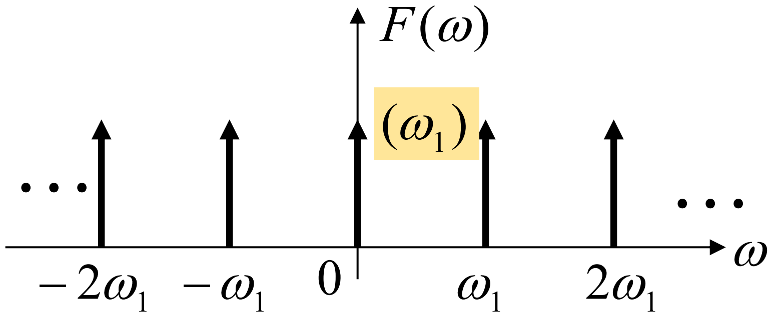
$$\delta_T(t) = \sum_{n=-\infty}^{\infty} \delta(t - nT_1)$$



$$F_n = \frac{1}{T_1}\int_{-T_1/2}^{T_1/2} \delta(t) e^{-jn\omega_1 t} dt = \frac{1}{T_1}\int_{-T_1/2}^{T_1/2} \delta(t) dt = \frac{1}{T_1}$$

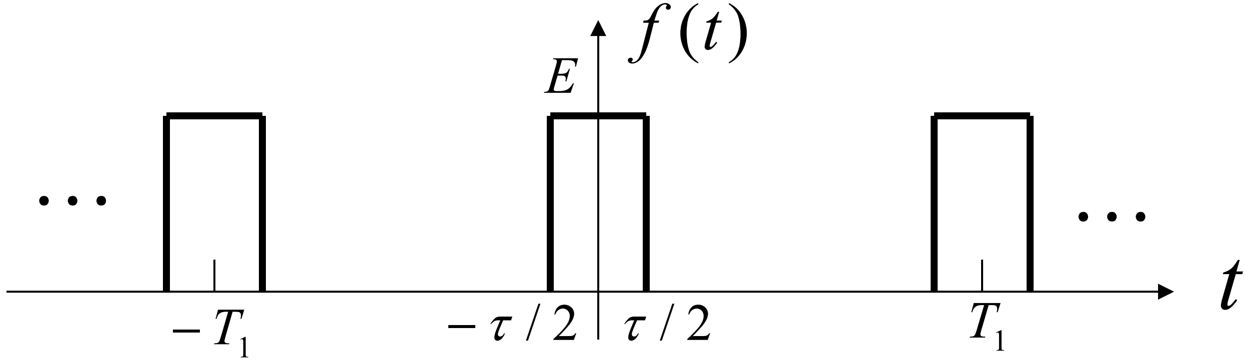
$$\delta_T(t) = \sum_{n=-\infty}^{\infty} F_n e^{jn\omega_1 t} = \frac{1}{T_1}\sum_{n=-\infty}^{\infty} e^{jn\omega_1 t}$$

$\times 2\pi$



$$F(\omega) = \frac{2\pi}{T_1}\sum_{n=-\infty}^{\infty} \delta(\omega - n\omega_1) = \omega_1 \sum_{n=-\infty}^{\infty} \delta(\omega - n\omega_1) \quad (2)$$

Example 3-10: Find the Fourier Series and Fourier Transform of the Periodic Rectangular Pulse Signal. Given that the periodic rectangular pulse signal $f(t)$ has an amplitude E, pulse width τ, period T_1, and angular frequency $\omega_1 = 2\pi / T_1$.



Solution: The complex amplitude of the Fourier series is:

$$F_n = \frac{E\tau}{T_1}\mathrm{Sa}\left(\frac{n\omega_1\tau}{2}\right)$$

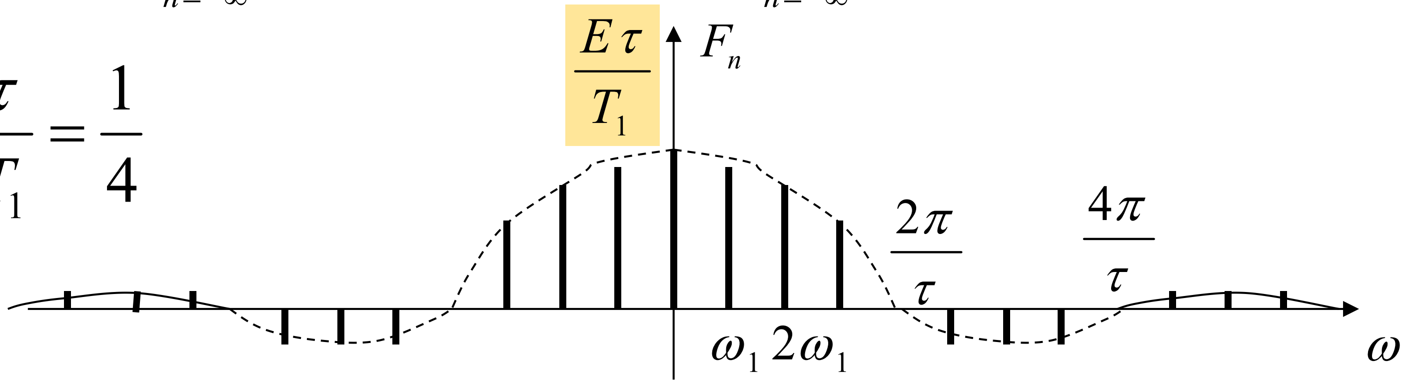
$$f(t) = \sum_{n=-\infty}^{\infty} F_n e^{jn\omega_1 t} = \frac{E\tau}{T_1}\sum_{n=-\infty}^{\infty}\mathrm{Sa}\left(\frac{n\omega_1\tau}{2}\right)e^{jn\omega_1 t}$$

3.9 Fourier Transform of Periodic Signals

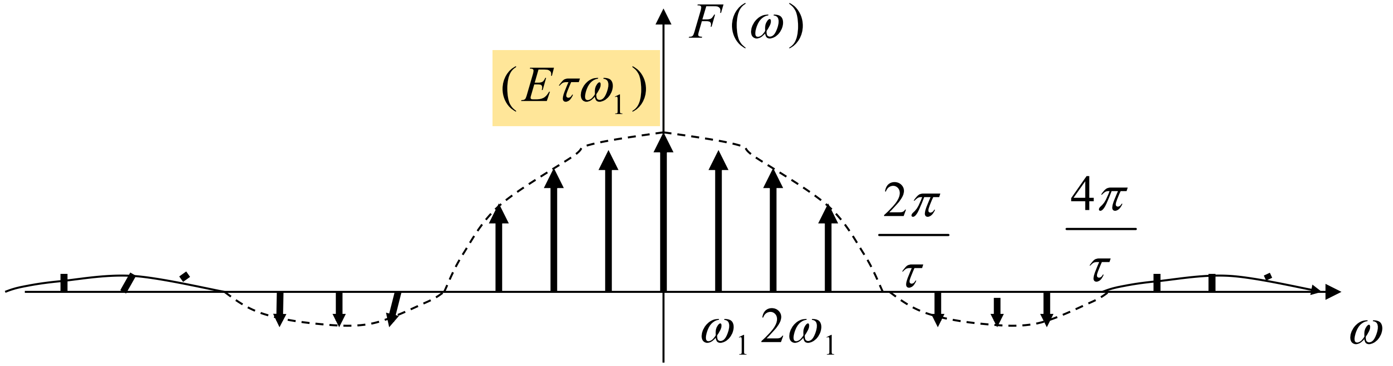
$$f(t)=\sum_{n=-\infty}^{\infty}F_n e^{jn\omega_1 t}=\frac{E\tau}{T_1}\sum_{n=-\infty}^{\infty}\mathrm{Sa}\left(\frac{n\omega_1\tau}{2}\right)e^{jn\omega_1 t}$$

$$F(\omega)=2\pi\sum_{n=-\infty}^{\infty}F_n\delta(\omega-n\omega_1)=E\tau\omega_1\sum_{n=-\infty}^{\infty}\mathrm{Sa}\left(\frac{n\omega_1\tau}{2}\right)\delta(\omega-n\omega_1)$$

E.g., $\dfrac{\tau}{T_1}=\dfrac{1}{4}$



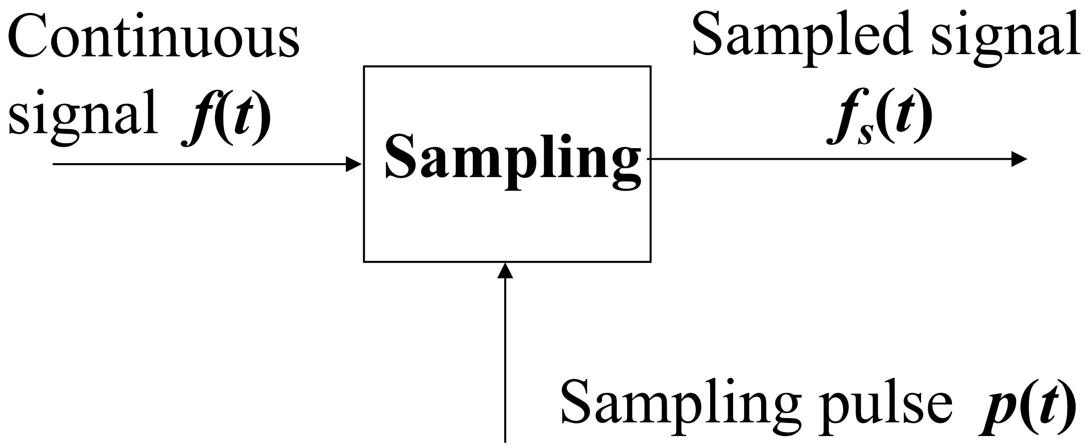
$\times 2\pi$



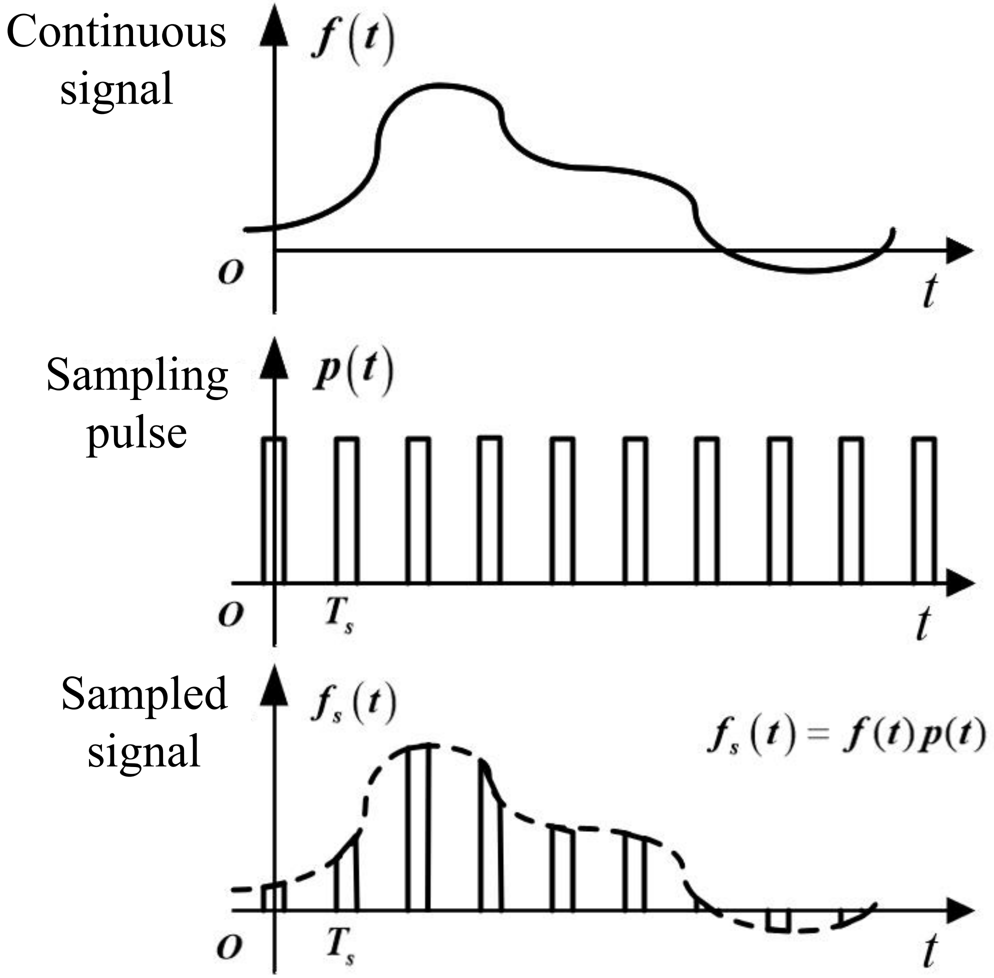
3.10.1 Sampling of Signals

Sampling -- Use a sampling pulse sequence to "sample" a series of discrete samples from a continuous signal.

This discrete signal is called a **sampled signal.**



T_s— **Sampling Period**

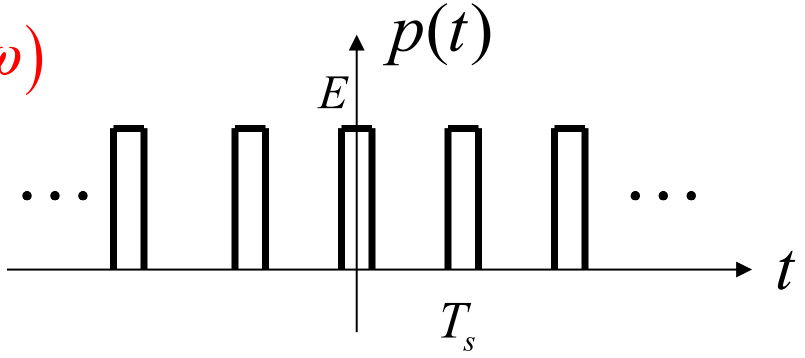


3.10.2 Fourier Transform of Sampled Signals

Let the Fourier transform of the continuous signal $f(t)$ be $F(\omega)$

the Fourier transform of the sampling pulse $p(t)$ be $P(\omega)$

the Fourier transform of the sampled signal $f_s(t)$ be $F_s(\omega)$



$$P(\omega) = 2\pi \sum_{n=-\infty}^{\infty} P_n \delta(\omega - n\omega_s)$$

$$\omega_s = 2\pi f_s = \frac{2\pi}{T_s}$$

Sampling Angular Frequency (ω_s)　**Sampling Frequency** (f_s)

where: $$P_n = \frac{1}{T_s}\int_{-\frac{T_s}{2}}^{\frac{T_s}{2}} p(t) e^{-jn\omega_s t} dt$$

$$f_s(t) = f(t)p(t)$$

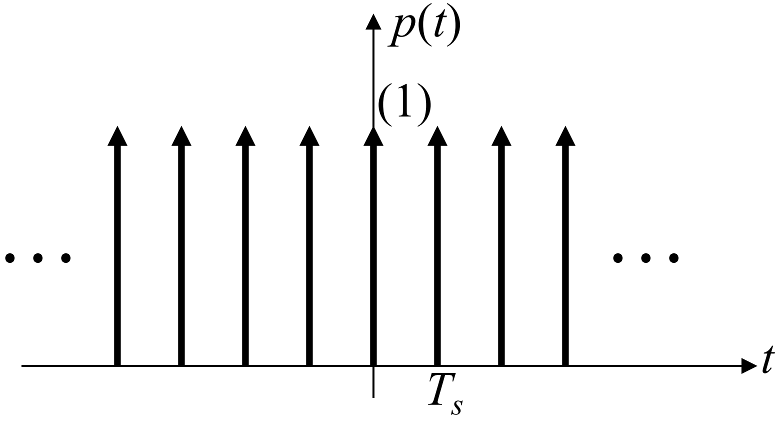
$$F_s(\omega) = \frac{1}{2\pi} F(\omega) * P(\omega) = \frac{1}{2\pi} F(\omega) * 2\pi \sum_{n=-\infty}^{\infty} P_n \delta(\omega - n\omega_s)$$

Frequency-domain Convolution Theorem

so $$F_s(\omega) = \sum_{n=-\infty}^{\infty} P_n \cdot F[(\omega - n\omega_s)]$$

Impulse Sampling

If the sampling pulse $p(t)$ is an impulse sequence, this is called "**impulse sampling**" or "**ideal sampling**".



$$p(t) = \delta_T(t) = \sum_{n=-\infty}^{\infty} \delta(t - nT_s)$$

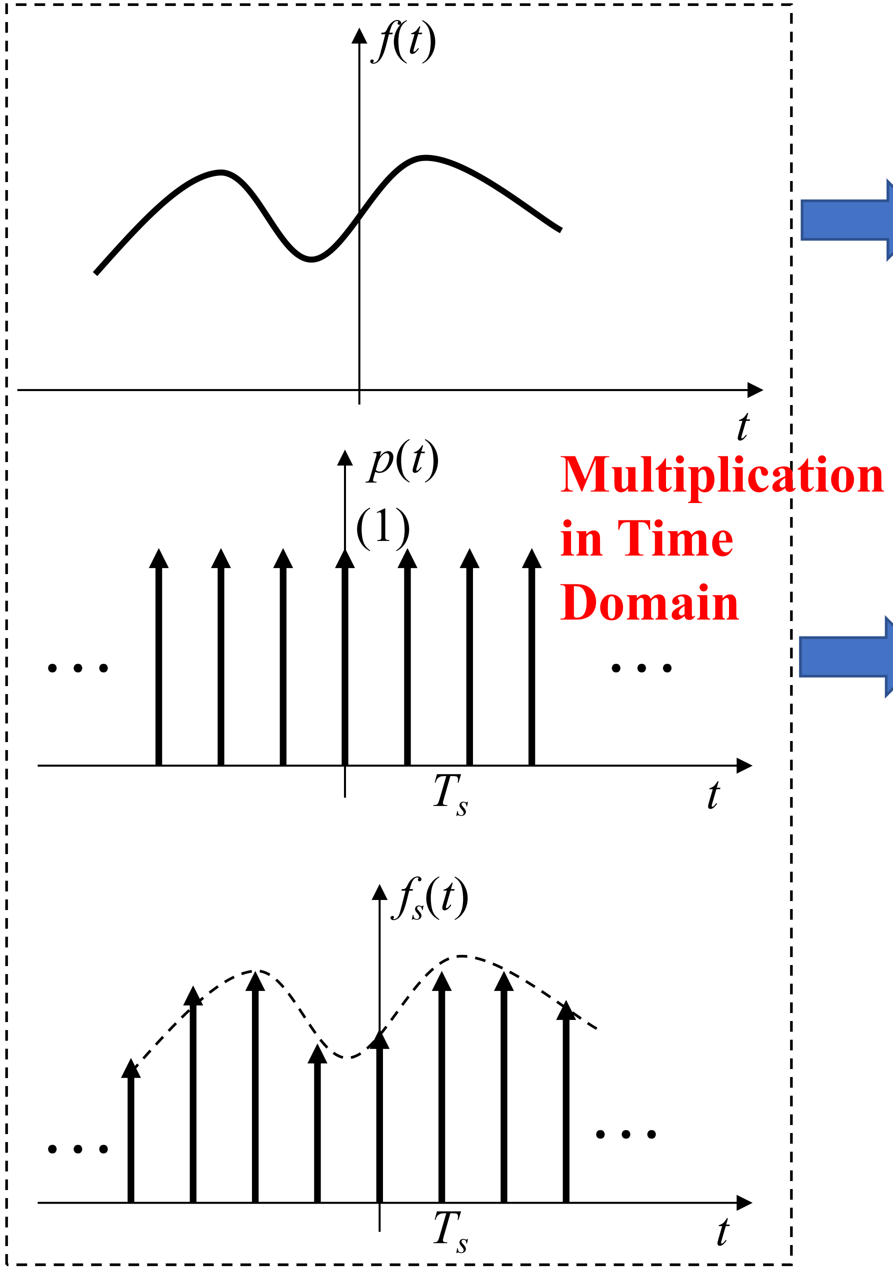
$$F_s(\omega) = \sum_{n=-\infty}^{\infty} P_n \cdot F(\omega - n\omega_s)$$

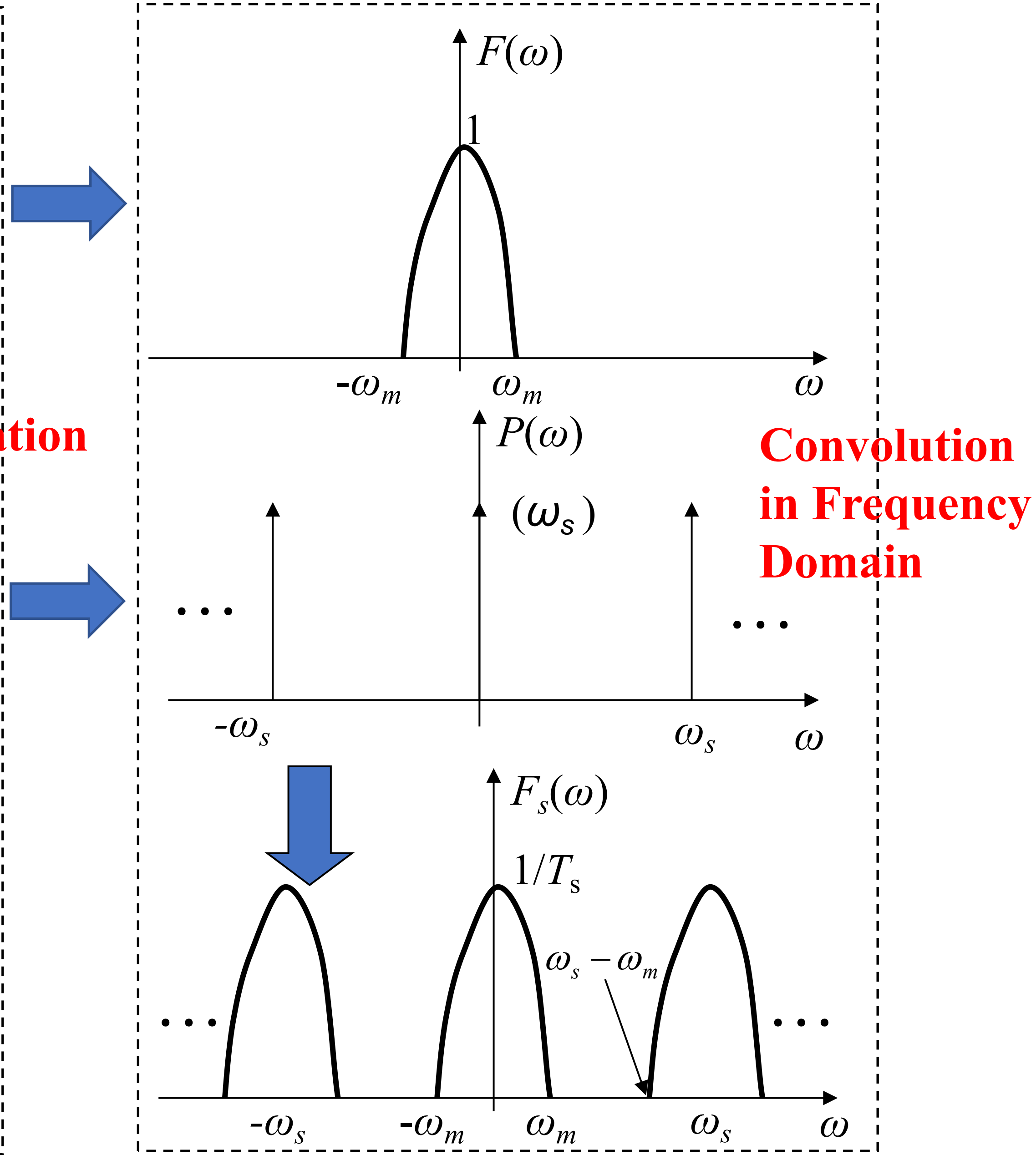
$$= \frac{1}{T_s} \sum_{n=-\infty}^{\infty} F(\omega - n\omega_s)$$

Periodically extended in the frequency domain with period ω_s

where $P_n = \frac{1}{T_s} \int_{-\frac{T_s}{2}}^{\frac{T_s}{2}} \delta(t) e^{-jn\omega_s t} dt = \frac{1}{T_s}$

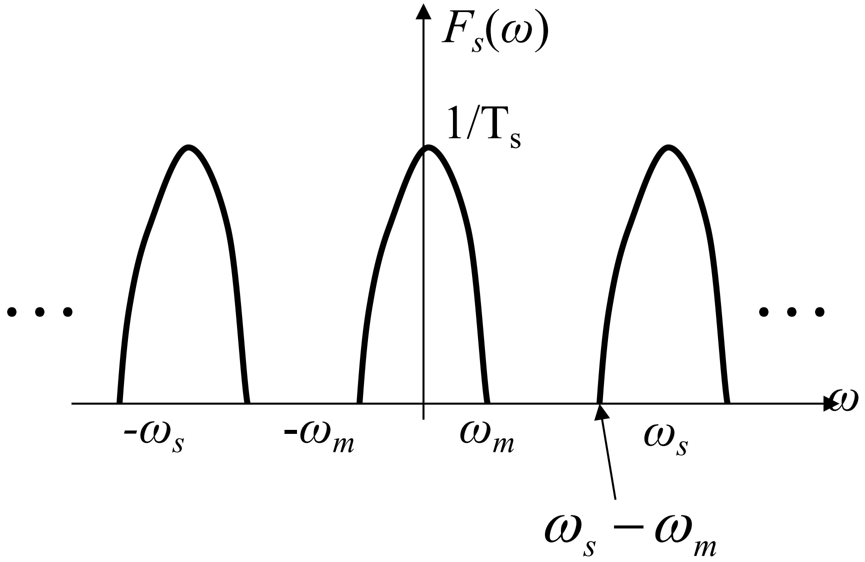
3.10 Fourier Transform of Sampled Signals





3.11 Sampling Theorem

When sampling a continuous signal with sampling pulses, **how to select the sampling frequency and sampling period**? And **how to recover the original continuous signal from the sampled signal**?

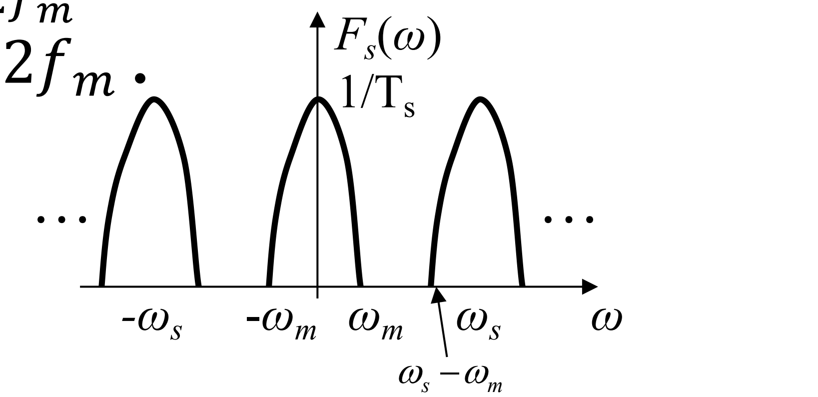


It can be seen from the above figure that only when $\omega_s \geq 2\omega_m$ is satisfied, will $F_s(\omega)$ not produce spectral aliasing, that is, $f_s(t)$ retains all the information of the original continuous-time signal.

At this time, as long as $f_s(t)$ is applied to the **ideal low-pass filter**, the original signal $f(t)$ can be recovered.

3.11 Sampling Theorem

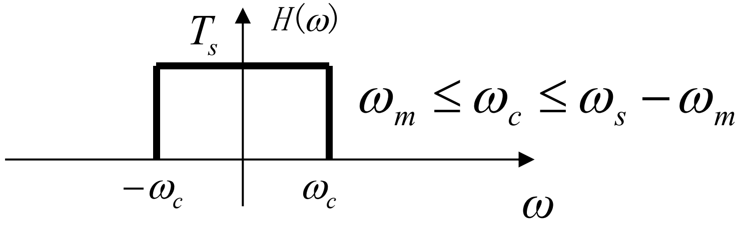
Time-domain Sampling Theorem: A band-limited signal $f(t)$, if its spectrum is confined to the range $-\omega_m \sim \omega_m$, can be uniquely represented by its equally spaced samples when the sampling interval is no larger than $\frac{1}{2f_m}$ (where $\omega_m = 2\pi f_m$) or the sampling frequency is greater than or equal to $2f_m$.



Nyquist Sampling Frequency: The minimum allowable sampling frequency.

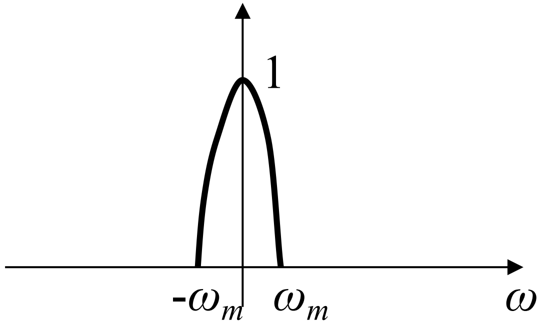
$$\boxed{f_{s\min} = 2f_m}$$

Frequency Response of the Ideal Low-pass Filter

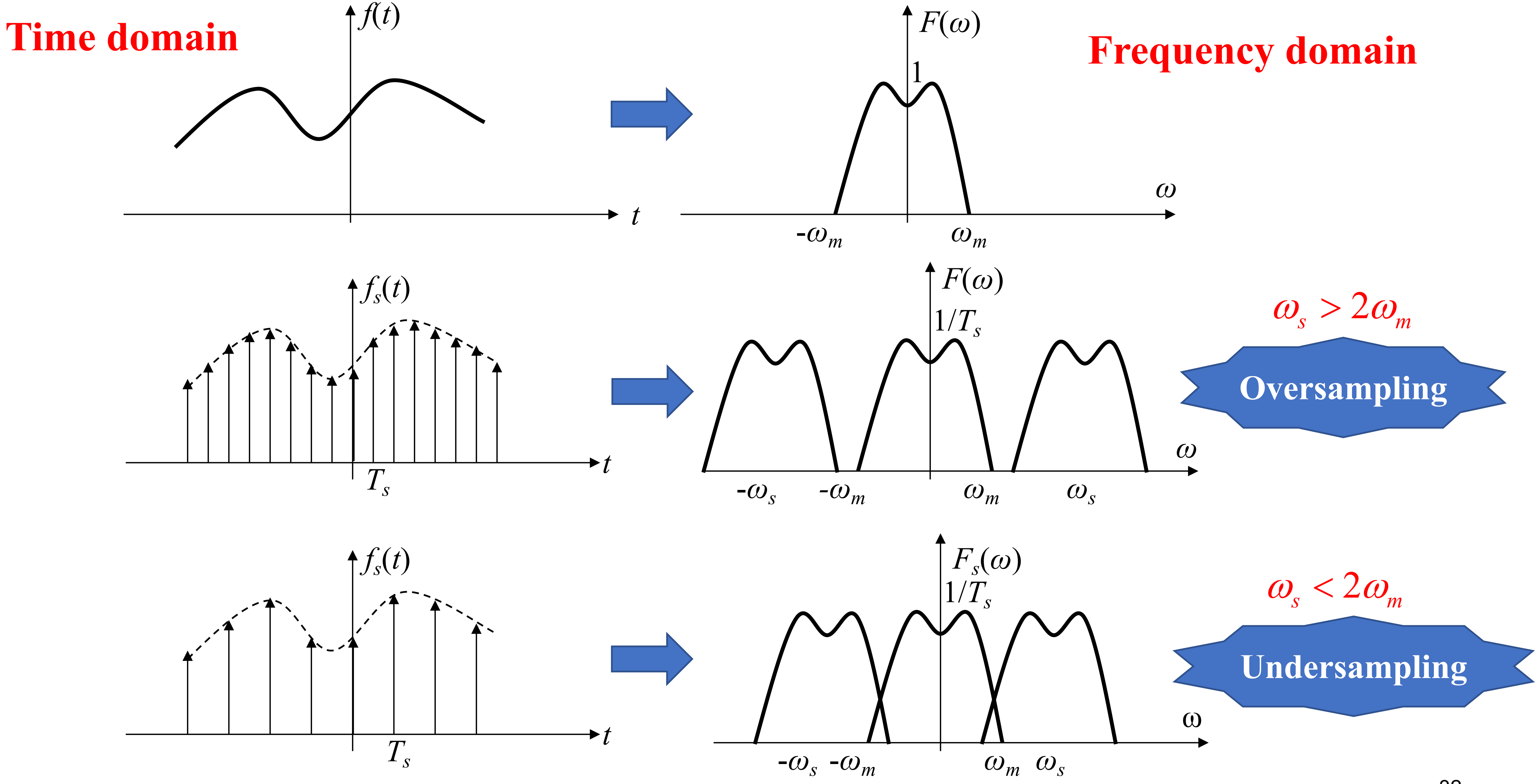


Nyquist Sampling Interval: The maximum allowable sampling interval.

$$\boxed{T_{s\max} = \frac{1}{f_{s\min}} = \frac{1}{2f_m}}$$



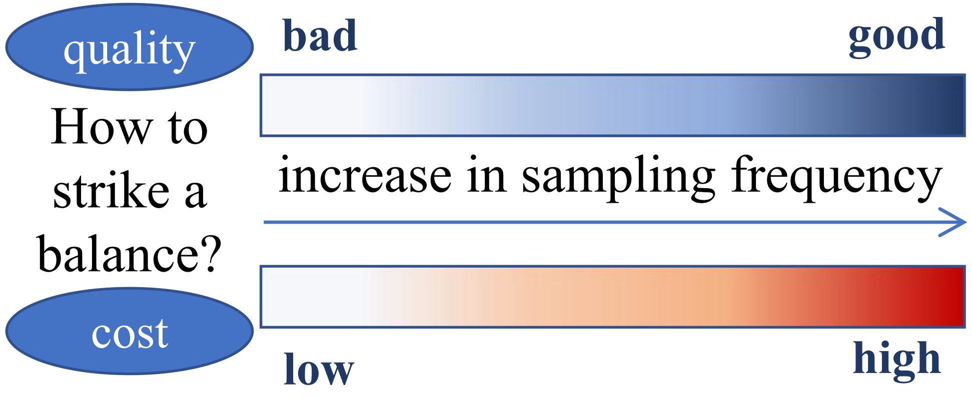
3.11 Sampling Theorem



3.11 Sampling Theorem

Example 3-11： CDs usually adopt a sampling frequency of 44 kHz. Please explain the reason. Is a higher sampling frequency always better?

- Most human ears can perceive frequencies ranging from 20 Hz to 16 kHz. To test your audible frequency range, you can visit: https://www.bilibili.com/video/av40724545/
- The corresponding Nyquist sampling frequency is 32 kHz.
- A pre-alias filter can be used to remove frequency components above 16 kHz.
- For high-quality audio signals, we generally use sampling frequencies higher than the Nyquist sampling frequency, such as 44 kHz, 48 kHz, 96 kHz, or 128 kHz.
- The selection of sampling frequency needs to strike a balance between sound quality and cost.



Selection of Sampling Frequency in Engineering

- If the highest frequency ω_m of the signal is known, generally take $\omega_s \geq (5 \sim 10)\omega_m$
- If the bandwidth ω_B of the signal is known, generally take $\omega_s \geq (5 \sim 10)\omega_B$
- If the highest frequency of the interference signal is ω_f, it should satisfy $\omega_s \geq 2\omega_f$

Empirical Data of Sampling Period in Industrial Control Systems:

Flow signal: $T_s = 1 \sim 5$ seconds;

Pressure signal: $T_s = 3 \sim 10$ seconds;

Liquid level signal: $T_s = 6 \sim 8$ seconds ;

Temperature signal: $T_s = 15 \sim 20$ seconds

Component signal: $T_s = 15 \sim 20$ seconds

Frequency-domain Sampling Theorem: If the signal $f(t)$ is a time-limited signal concentrated in the time range $-t_m \sim +t_m$, and if its spectrum $F(\omega)$ is sampled in the frequency domain with a frequency interval no larger than $\frac{1}{2t_m}$, then the sampled spectrum $F_s(\omega)$ can uniquely represent the original signal.

Example 3-12: Given the signal $f(t)=\mathrm{Sa}(2t)$, sample it with $\delta_T(t)=\sum_{n=-\infty}^{\infty}\delta(t-nT_s)$,

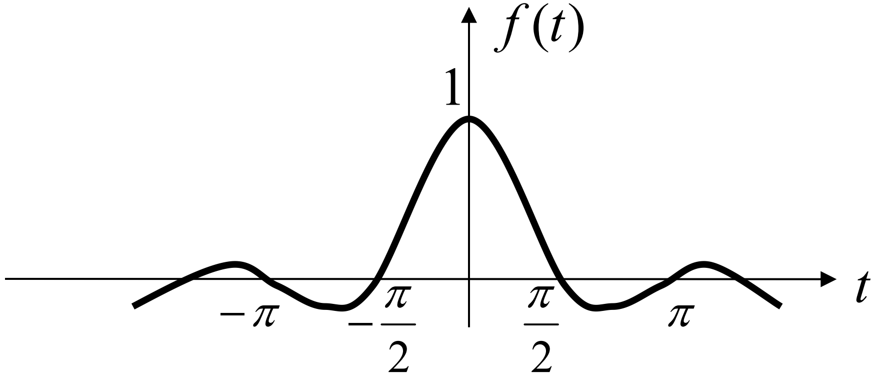
(1) Determine the Nyquist sampling frequency;

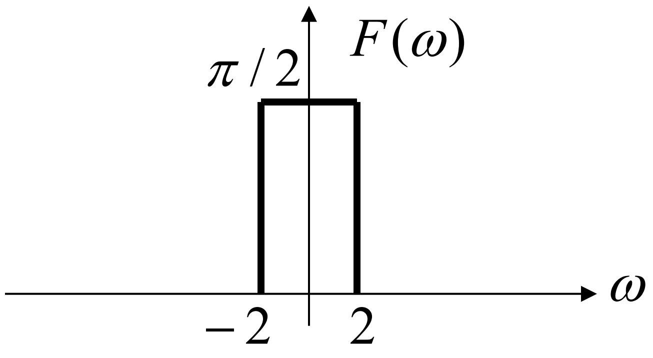
(2) If $\omega_s=6\omega_m$, find the sampled signal $f_s(t)=f(t)\delta_T(t)$, and draw its waveform;

(3) Find $F_s(\omega)=\mathcal{F}[f_s(t)]$, and draw its spectrum;

(4) Determine the cutoff frequency ω_c of the low-pass filter.

Solution: (1) $\because f(t)=\mathrm{Sa}(2t)$ $\quad\therefore F(\omega)=\dfrac{\pi}{2}[u(\omega+2)-u(\omega-2)]$





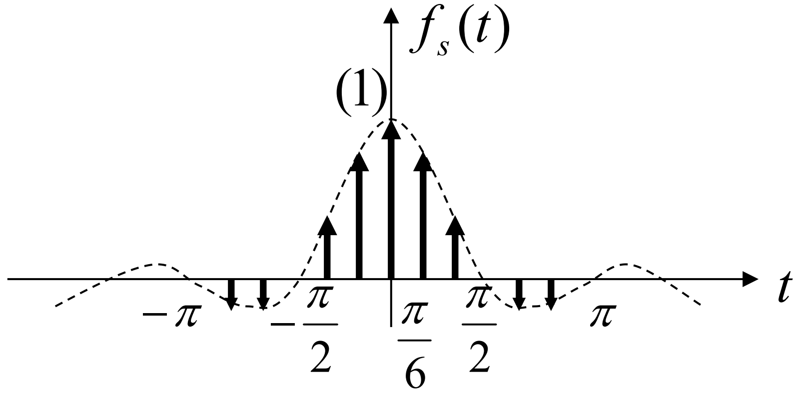
Nyquist sampling frequency:

$$f_{s\ \min}=2f_m=2\times\frac{2}{2\pi}=\frac{2}{\pi}\ \text{Hz}$$

（2）$\because \omega_s = 6\omega_m = 12\ \text{rad/s} \qquad \therefore T_s = \dfrac{2\pi}{\omega_s} = \dfrac{2\pi}{12} = \dfrac{\pi}{6}\ \text{s}$

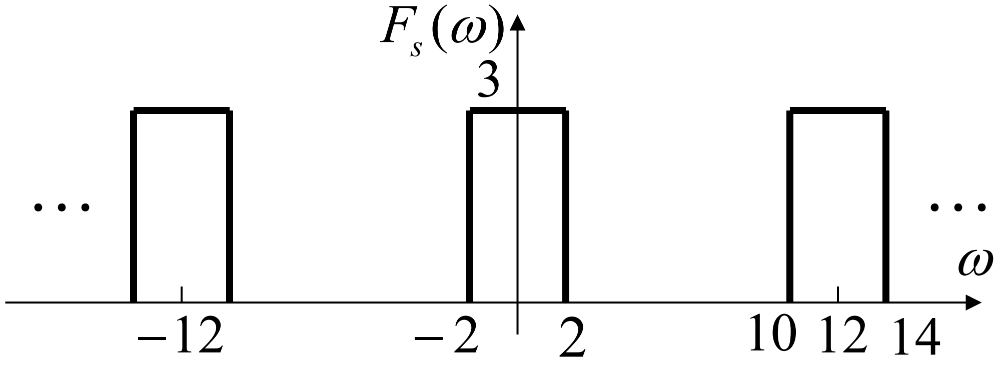
$$f_s(t) = f(t)\delta_T(t) = \sum_{n=-\infty}^{\infty} f(nT_s)\delta(t-nT_s)$$

$$= \sum_{n=-\infty}^{\infty} \text{Sa}(2t)\Big|_{t=nT_s}\delta(t-nT_s) = \sum_{n=-\infty}^{\infty} \text{Sa}\left(\frac{n\pi}{3}\right)\delta(t-nT_s)$$

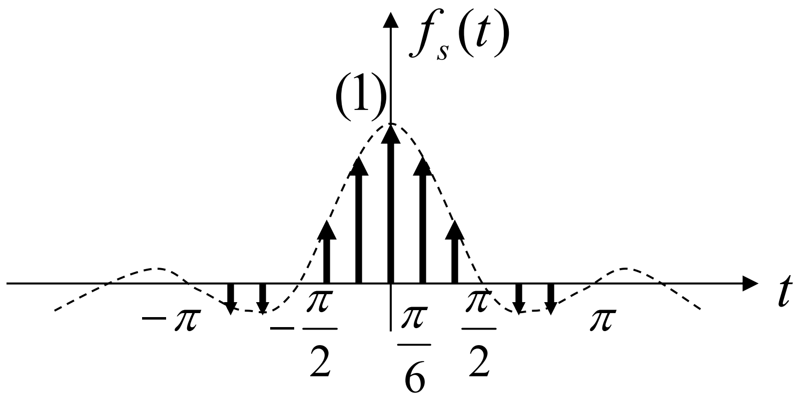


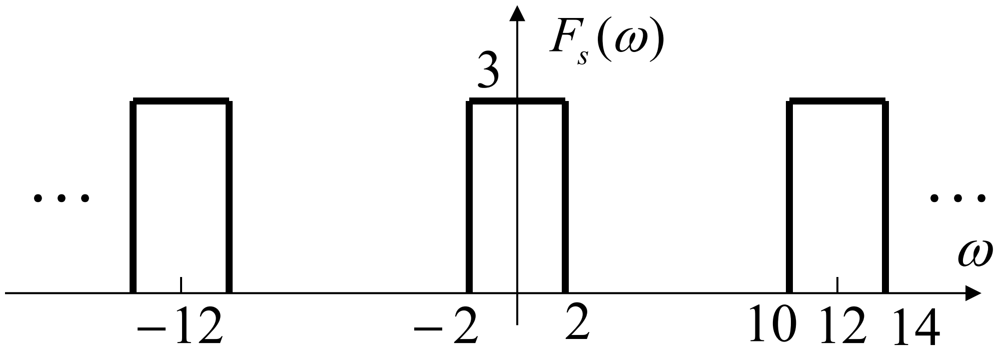
（3）$F_s(\omega) = \dfrac{1}{T_s}\displaystyle\sum_{n=-\infty}^{\infty} F(\omega - n\omega_s) = \dfrac{6}{\pi}\displaystyle\sum_{n=-\infty}^{\infty} F(\omega - 12n)$

$$= 3\sum_{n=-\infty}^{\infty}[u(\omega+2-12n) - u(\omega-2-12n)]$$



(4)

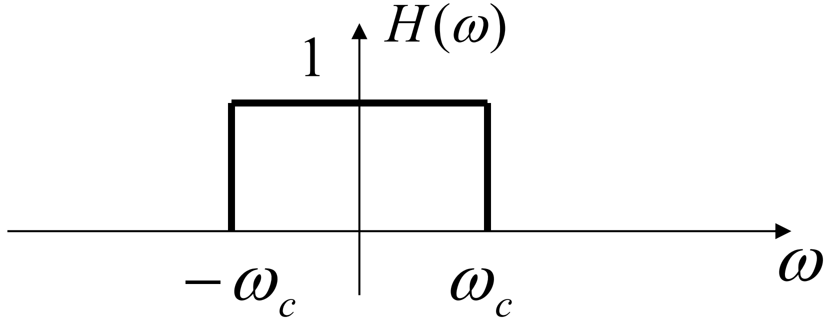




The cutoff frequency ω_c of the low-pass filter should satisfy the following formula:

$$\omega_m \le \omega_c \le \omega_s - \omega_m$$

then $2 \le \omega_c \le 10$



Homework

- Repeat the examples in the lecture notes.
- 3.58(g), 3.59(f), 4.16(d) and 4.29(a) in the textbook.